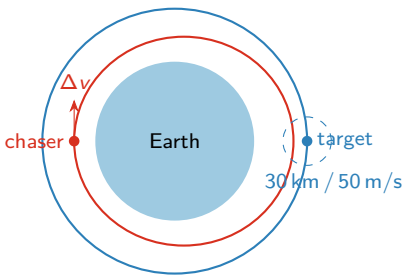
# Learning Orbital Rendezvous

## A reinforcement-learned guidance policy for a chaser spacecraft

Peter Brown

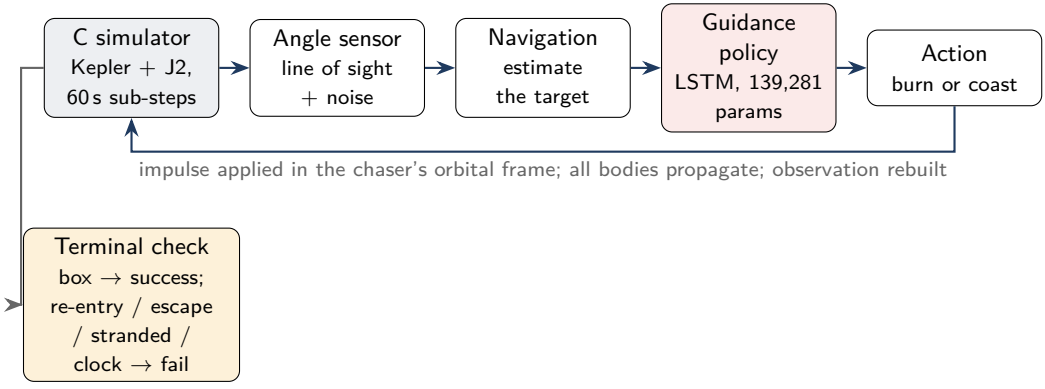
April–August 2026

# Goal: maneuver a chaser into a box around a passive target



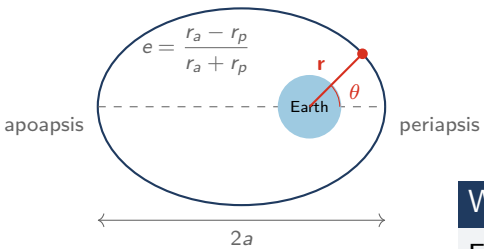

- **Success** inside 30 km and 50 m/s of the target
- **Failure** re-enter, escape, run dry, or run out of clock
- **Tank** 478 m/s (15 % propellant); the generalist draws 353–656 m/s per episode
- **Start** random orbits, phase gap up to $\pm 180°$; LEO 300–800 km in the headline, up to 8,000 km and $e \leq 0.3$ in the generalist
- **Act** a fixed-size burn, or a coast of 1 min to 24 h

# One loop, every decision




- The policy sees 38 numbers and picks one of 31 actions. It never sees the target's true state in the angles-only configuration.
- The same loop runs in allocation-free C, action for action identical to PyTorch.

# Simulator: analytic two-body plus J2, written in C

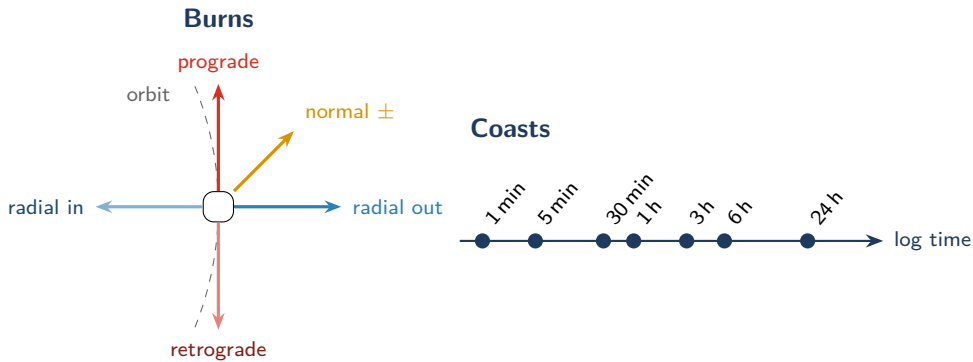



- **State** orbital elements $(a, e, i, \Omega, \omega, M)$
- **Propagate** Kepler's equation, closed form at any step
- **Burns** impulsive, in the chaser's own frame
- **J2** Earth's bulge slowly rotates the orbit plane

## Why analytic

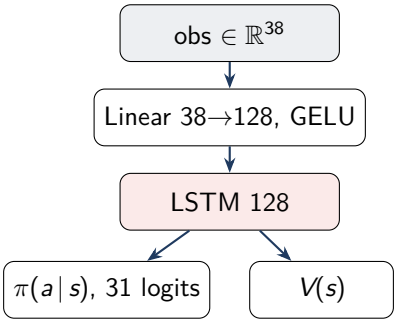
Exact at any step size, so a 24-hour coast is one solve. About half a million environment steps per second on one laptop CPU.

# Actions: a short menu of fixed burns and coasts



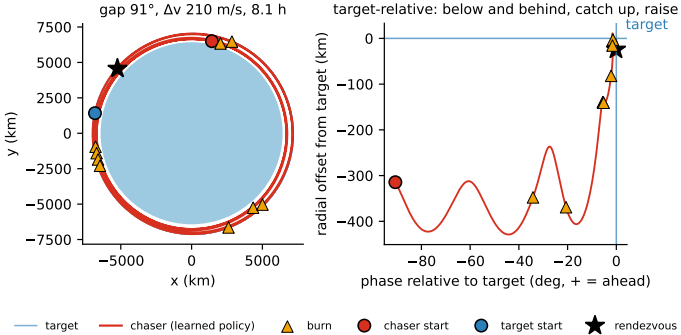

burn grades 1, 2, 5, 10, 25 m/s · 31 actions in all

# Policy and training



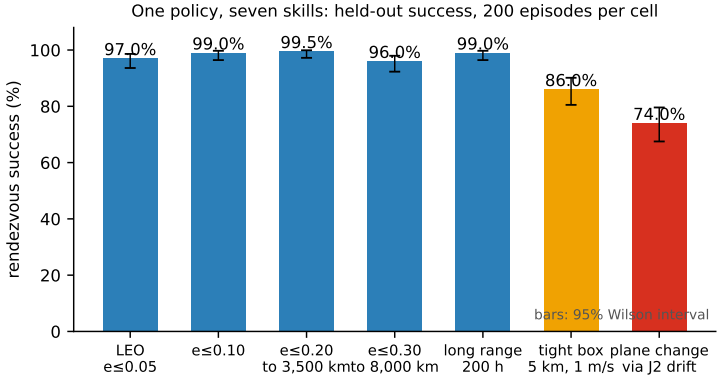

- **Small and recurrent** 139,281 parameters; the LSTM carries what the last burn did
- **PPO** 1024 parallel environments, 50 M steps per stage, about two hours each
- **Reward** $+1$ success, $-1$ failure, with potential-based shaping in between
- **Curriculum** one new difficulty per stage, warm-started

# What it learned, 2-D: phasing, shape matching, and when to wait



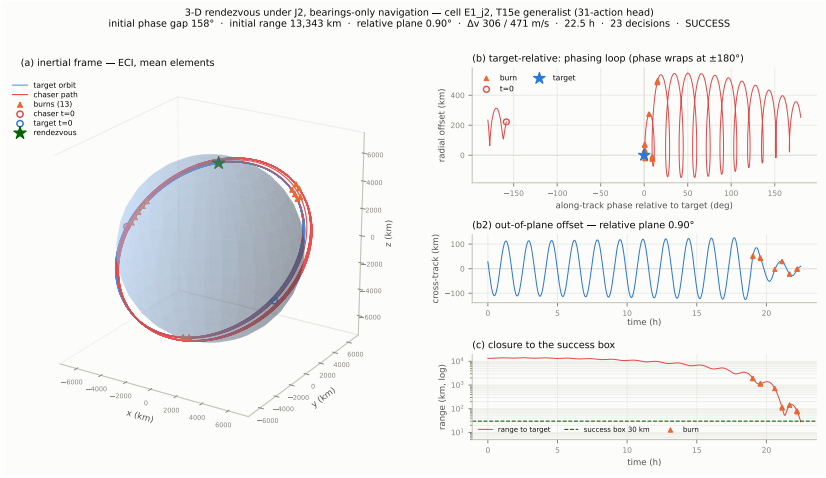

- **200/200** held-out episodes on **5/5 training seeds**; wide envelope (300–8,000 km, $e \leq 0.30$) 800/800 on 4 seeds.
- Fuel: median 1.18× a two-impulse optimum on the same clock (appendix A4).
- Phasing by altitude, apse and eccentricity matching, and coasting through windows where a burn would not help.

# One policy, seven rendezvous skills, in 3-D with J2



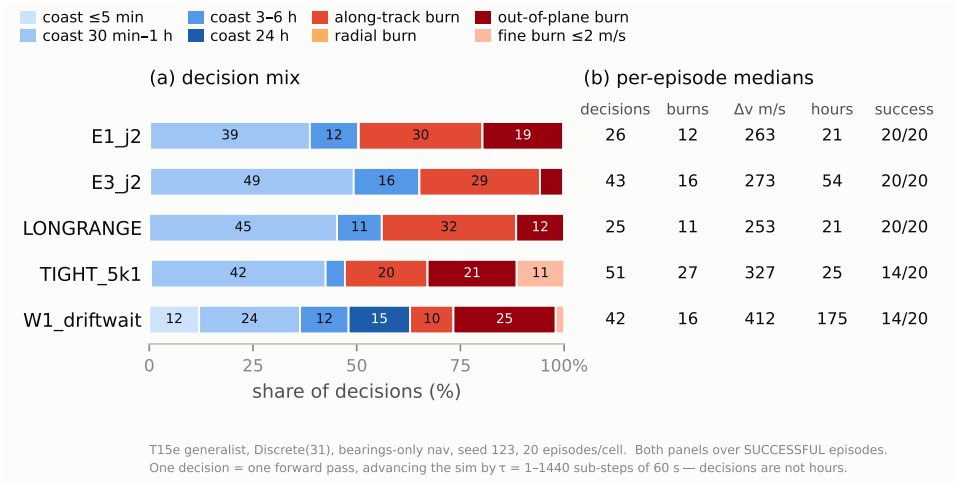

Same network, same weights, every cell. 200 held-out episodes per skill, flying on the angles-only estimate.

# The same policy flying one 3-D episode




A 158° gap and a 0.9° plane difference closed in 22.5 h with 306 m/s and 23 decisions, on the filter's estimate of the target.

# What the policy actually does, scenario by scenario




Decision mix of the seven-skill policy on five of its scenarios, 20 held-out episodes each (success rates on slide 8 are the 200-episode ones). The same weights coast a day at a time only when a plane gap can be drifted shut, use fine 1–2 m/s burns only for the tight box, and never touch the radial burns.

# What is open, and what I would do next

**Open**

- The seventh skill's final refinement chain is one training seed
- Fuel is $1.18\times$ optimal; the burn grid is coarse
- Chaser navigation assumed perfect; no eclipses or sensor bias

**Next**

- Finer burn grades
- More seeds on the generalist
- A maneuvering target

# Appendix

observation vector · phasing and J2 · the seven cells · fuel audit · Lambert search · navigation · C port · the seventh skill · FAQ · verification · seeds · curriculum · the seventh skill · LSTM · extensions · PPO · fuel awareness

# A1 Observation vector (38 floats, roughly in $[-1, 1]$)

| slots | value | why |
|---|---|---|
| 0–1 | chaser altitude / scale, $e$ | size and shape |
| 2–3 | $\sin\theta, \cos\theta$ | position on orbit, no wrap |
| 4–5 | $v_r/v_{circ}$, $v_t/v_{circ}$ | radial and tangential speed |
| 6 | fuel fraction | budget remaining |
| 7–8 | target altitude, $e$ (estimated) | target size and shape |
| 9–12 | $\sin, \cos\ \omega$ for both | apse orientations |
| 13–14 | $\sin, \cos\ \Delta\lambda$ (mean-longitude gap) | the phase gap |
| 15 | (cap − step)/cap | episode clock |
| 16 | $\cos(\omega_s - \omega_t)$ | apsidal alignment |
| 17–20 | nearest body: distance, bearing, closing rate, keep-out | Earth |
| 21–25 | relative-inclination vector; $\Delta\mathbf{e} \cdot \hat{h}_t$; cross-track offset and rate | 3-D block |
| 26–28 | $\Delta v_{plane}$, $\Delta v_{remaining}$, feasibility margin | fuel ledger |
| 29–30 | $\cos i_s$, $\cos i_t$ | J2 |
| 33–36 | LVLH radial / along-track offset; relative velocity, rotating-frame corrected | relative state |
| 37 | target mean motion | |

Target quantities are the filter's estimate in bearings-only mode. The policy is not given the covariance (tested: a $\sigma$ channel added nothing).

# A2 Phasing, and plane change by burning versus by J2

**Phasing** $T = 2\pi\sqrt{a^3/\mu}$; a chaser 100 km below a LEO target gains about 4° of phase per orbit, so "drop, wait $N$ orbits, raise" is the textbook catch-up and it is what the policy does.

**Plane change by burning** $\Delta v = 2v\sin(\Delta i/2)$. At 7.5 km/s a 3° change costs about 390 m/s, most of a 478 m/s tank.

**Plane change by J2** Earth's equatorial bulge makes the node line drift around the pole at

$$\dot{\Omega} = -\tfrac{3}{2}\, J_2\, n \left(\frac{R_E}{p}\right)^2 \cos i, \qquad p = a(1-e^2).$$

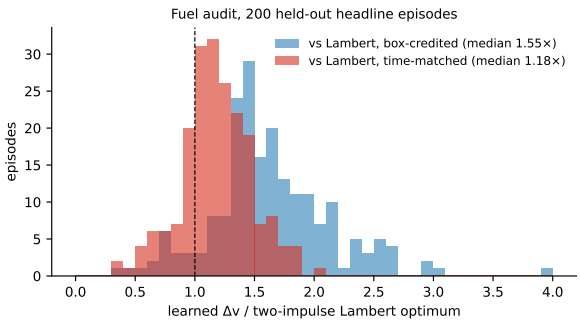
The rate depends on altitude and inclination, so two orbits at different altitudes drift apart in $\Omega$ at a controllable relative rate. Dip low, coast a day at a time, let the node drift close the plane gap, then raise. In the specialist about half of the plane alignment was bought this way.

# A3 The seven cells

| cell | altitude band (km) | $e_{max}$ | box | clock (sub-steps) | success |
|---|---|---|---|---|---|
| E0_j2 | 300–800 | 0.05 | 30 km / 50 m/s | 3,000 | 97.0 % |
| E1_j2 | 300–1,500 | 0.10 | 30 km / 50 m/s | 3,000 | 99.0 % |
| E2_j2 | 300–3,500 | 0.20 | 30 km / 50 m/s | 4,500 | 99.5 % |
| E3_j2 | 300–8,000 | 0.30 | 30 km / 50 m/s | 6,000 | 96.0 % |
| LONGRANGE | 300–3,500 | 0.12 | 30 km / 50 m/s | 12,000 | 99.0 % |
| TIGHT_5k1 | 300–800 | 0.05 | 5 km / 1 m/s | 3,000 | 86.0 %† |
| W1_driftwait | 300–800 | 0.05 | 30 km / 50 m/s | 22,000 | 74.0 % |

†92.5 % at the training fuel floor. All cells: 3-D, J2 on, targets inclined 30–60°, relative plane up to 0.75–1° (2–5° for W1), bearings-only navigation, Discrete(31) head, fuel budget sampled per episode (353–656 m/s). Held-out seed 123, 200 episodes per cell, greedy actions, LSTM state zeroed per episode. A second training seed reproduces six of the seven cells (A11).

# A4 Fuel audit against a two-impulse optimum



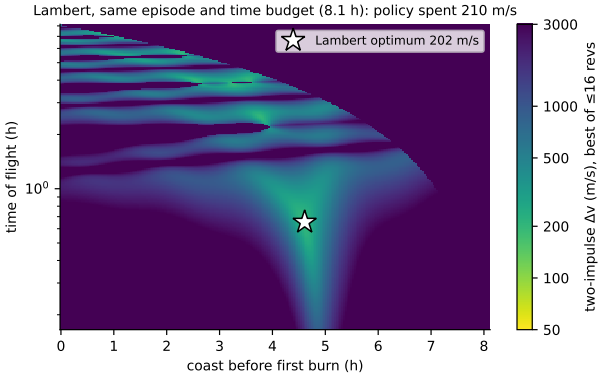

For each episode an offline solver finds the cheapest two-burn transfer from the chaser's start to the target, searching over how long to wait before the first burn and how long to fly, under the same clock the policy used.

**Time-matched** must hit the target's exact position and velocity: median learned $\Delta v$ is **1.18**× it.

**Box-credited** the same solution minus the 50 m/s velocity tolerance the policy is never asked to close: **1.55**×. The fairer of the two.

200 held-out episodes. Ratios below 1 happen because arrival is credited inside a box and a two-burn plan is not the unconstrained optimum.

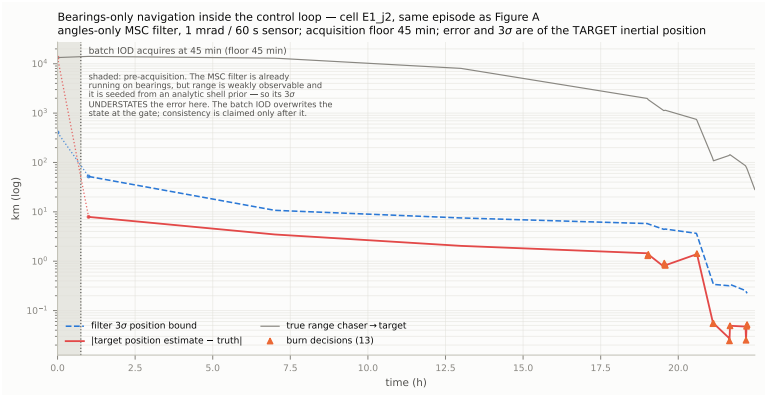
# A5 What the Lambert search looks like for one episode




Every cell is one candidate plan: wait this long, then fly for that long. Colour is the two-burn $\Delta v$ of the best conic connecting the two points in that time, including multi-revolution solutions. The banded structure is the orbital period showing through: cheap plans recur once per orbit.

The star is the cheapest plan inside the policy's own clock. On this episode the optimum was 202 m/s and the policy spent 210.

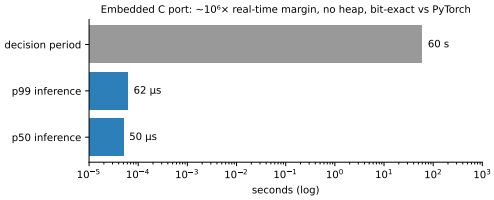
# A6 Navigation in one picture: blind, acquire, converge




The chaser measures only the direction to the target, not the distance. One direction tells you nothing about range; a sequence of directions while both orbits curve does. A batch solver tries many ranges along the first line of sight, propagates each, and keeps the one whose predicted directions match what was seen. That hands a full state to a recursive filter that keeps refining it.

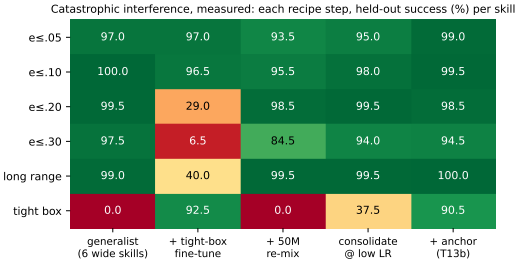
Policy result: trained on truth and handed the estimate, **51 %**; trained with the filter in the loop, **97 %** (2-D tight box) and **46→98 %** across three seeds in 3-D. It learned to wait for acquisition and burn nothing while blind (median $\Delta v$ before acquisition 75→0 m/s).

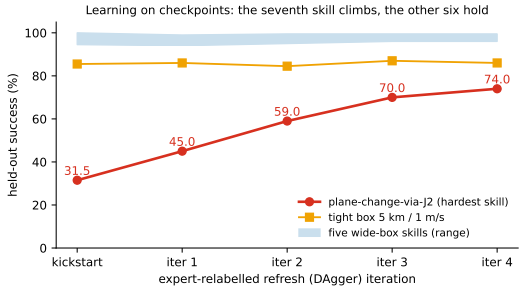
# A7 C port of the policy




- The trained weights exported to a C header; forward pass written by hand: linear, GELU, LSTM cell, two linear heads.
- No heap, no globals; the caller owns the 1 kB of recurrent state; fixed work per call.
- Checked against PyTorch: identical action on 7,048 recorded observations and an identical action stream over 200 closed-loop episodes.
- Median 50 $\mu$s per decision on a laptop against a 60 s decision period (a benchmark, not a worst-case timing).

# A8 How the seventh skill was trained




- Left: fine-tuning on one skill damaged others; a low learning rate plus replay of each skill's own trajectories held all six.
- Right: the plane-change skill never bootstrapped from reward. A scripted expert demonstrated it, the policy was refreshed on the expert's corrections four times, and each refresh was scored on 200 held-out episodes.

# A9 FAQ

**Why RL, when Lambert exists?** Lambert needs perfect state and gives an open-loop two-burn plan; the policy is closed-loop on an estimate, respects a budget, and one network covers seven regimes. Lambert is the grader, not the competitor.

**Why discrete actions?** PPO is stable with them; a continuous head collapsed early on. The grid costs fuel (1.14–1.18× across two evaluation sets); finer grades are the obvious next step.

**Is 200 episodes enough?** At 97 % the 95 % Wilson interval is about ±2.5 pp; for 0/200 failures the lower bound is 98.2 %. I report intervals, not "100 %".

**Single seed?** The six-skill policy reproduced at a second training seed (tight box 92.5 %, wide cells 93.5–100 %). The seventh skill's four-step refinement was run once by design; its first two steps reproduced at a second seed with matching gains. The 2-D headline (5), wide envelope (4), 3-D nav and J2 rungs (3) are multi-seed.

**Why coast 24 h?** J2 nodal precession closes a plane gap for free; burning it would exceed the tank. The policy learned to wait.

**How does the filter work?** Angles only, so range comes from how the bearing changes while the orbits curve; a batch solver picks the range that explains the first hour of bearings, then a recursive filter refines it. The finding I own is on the policy side: train on the estimate or lose 46 points.

**What would you do differently?** Audit against physics from day one; log learning curves on every run; design the observation with the filter in mind from the start.

# A10 Verification measures

| what | how | result |
|---|---|---|
| Propagator | independent Lagrange $f,g$ oracle sharing no code; 980 adversarial episodes, 618k sub-steps | worst error 0.68 m (float32 logging floor) |
| Conservation | energy and angular momentum on every coast step | at float precision |
| Transfer $\Delta v$ | Hohmann 400→800 km and LEO→GEO vs analytic | 217.02 vs 217.02 m/s; 3856.7 vs 3856.7 |
| Elements ↔ Cartesian | 400 random round trips | median $3 \times 10^{-8}$ m |
| Regression | three bitwise trajectory anchors (action-stream hash must not move) | caught real drift twice |
| Metrics | mutation test: 14 seeded errors in the evaluation code | 14/14 caught |
| Evaluation | held-out episode seed, greedy actions, Wilson intervals; 0/200 failures reported as $\geq$ 98.2 % | every number on these slides |
| Comparator | two-impulse Lambert optimum per episode, same clock | 1.18× median |
| Filter | NEES / NIS in-bounds fractions (range/bearing EKF) | 94 / 95 % |
| C port | action agreement on 7,048 observations; 200 closed-loop episodes | 100 %; identical action stream |

Sources: `T3_RECOVERY_CAMPAIGN.md` §4–7, `PROGRESS.md` physics table, `scripts/orbital/embedded/results/`.

# A11 Seeds and sample sizes

| result | training seeds | held-out episodes | note |
|---|---|---|---|
| 2-D headline, 30 km / 50 m/s | 5 | 200 per seed (+500 at a second eval seed) | 200/200 every seed |
| 2-D wide envelope (300–8,000 km, $e \leq 0.30$) | 4 | 200 per seed | 800/800 |
| 3-D rendezvous, truth state | 3 | 200 per seed | 200/200 every seed |
| 3-D bearings-only, tight box | 3 | 200 per seed | 589/600 pooled |
| J2 tight box, truth state | 3 | 200 per seed | 595/600 pooled |
| Six-skill generalist (anchored) | 2 | 200 per cell | seed 7: tight 92.5, wides 93.5–100 |
| Seventh skill, refinement chain | 1 (steps 1–2 at a second seed) | 200 per cell | 74.0 at seed 42; 47.5 after two steps at seed 7 |
| 2-D policy-on-estimate, 51→97 | 1 | 200 | replicated in 3-D across 3 seeds |

Held-out means a fresh evaluation seed never used in training. 200 episodes at 97 % gives a 95 % Wilson interval of about ±2.4 points.

# A12 The curriculum: one new difficulty per stage, warm-started

| stage | what changes | held-out result | cost |
|---|---|---|---|
| L1 | circular orbits, full $\pm 180^\circ$ phase gap, fresh weights | 200/200 | 50 M steps ($\sim$2 h) |
| L2 | eccentricity $\leq 0.05$, independent apse orientation | 200/200 | 50 M |
| WL3, WL4 | altitude to 2,000 then 8,000 km, $e$ to 0.30 | 200/200 each | 50 M each |
| 3-D | inclined planes, out-of-plane burns, 30-action head | 200/200 (3 seeds) | 50 M |
| nav | filter estimate replaces truth in the observation | 97–98 % | 50 M |
| J2 | secular drift on, tight box via three sub-rungs | 99.2 % (3 seeds) | $3 \times 50$ M |
| mixture | seven cells sampled per episode, tank sampled | slide 8 | 50–200 M |

- **Why one variable per stage** fresh training at the hard endpoint flat-lined six times; each stage starts from a policy that already solves the previous one, so the new difficulty is the only thing left to learn.
- **How a stage was declared done** a pre-registered gate ($\geq 60$ % held-out, 3 seeds); in practice every rung cleared 200/200 before moving on. By the number, not by eye.
- **Warm start** weights copied; new action-head rows zero-initialised so behaviour is unchanged at step zero.

# A13 Why reward never found the seventh skill

**The skill** close a 2–5° plane gap by parking at a different altitude and letting J2 nodal precession drift the planes together over days, then rendezvous.

**What reward offers** $+1$ at the end of a 175-hour episode, $-1$ otherwise, a shaping hint that cannot see the drift because it rewards closing the gap, not setting up the conditions that will close it later.

**What exploration offers** a random policy must choose the day-long coast repeatedly, at the right altitude, with no intermediate signal. The probability is vanishing; 0/200 under every reward variant tried.

**The fix** a scripted expert that knows the manoeuvre (91 % on the cell) demonstrates it; the policy is trained on those demonstrations (kickstart, 31.5 %), then refreshed four times on the expert's corrections of the states the policy itself visits (DAgger): 45, 59, 70, **74 %**.

**Why DAgger and not plain imitation** a policy that only sees expert states drifts into states the expert never showed it and has no idea what to do; relabelling its own visited states closes that gap.

**The framing** an exploration problem, not a capacity problem: the same 128-unit network holds the skill once shown it.

# A14 Why an LSTM, and what the memory carries

**Partial observability** the policy sees the filter's estimate of the target, not the truth. Two things live only in history:

- what the last burn did (the observation shows the state after, not the delta)
- where the filter is in its convergence (is the estimate still moving, or settled)

**Architecture** Linear 38→128, GELU, LSTM 128, two linear heads. 139,281 parameters. Hidden and cell state (256 numbers) persist across decisions within an episode and are zeroed at reset.

**Training detail** truncated backpropagation through time over 64 decisions; the recurrent state is carried across rollout boundaries.

**The obvious alternative, tested** give the policy the filter's covariance directly ($\sigma$ of range over range) as an extra observation channel.

| arm | native | truth |
|---|---|---|
| trained on estimate | 97.0 % | 97.0 % |
| same $+\sigma$ channel | 97.0 % | 98.5 % |

No change. The recurrent policy already infers what it needs about the estimate's quality from how the estimate moves.

200 episodes per arm, seed 123, 5 km / 1 m/s box. A memoryless control was never run; that ablation is on the list.

# A15 Extensions: a maneuvering target, and multiple agents

**Maneuvering target** the navigation stack changes before the policy does.

- today's filter assumes a ballistic target; an unmodelled burn shows up as a growing innovation and a diverging estimate
- first step: residual monitoring (a $\chi^2$ gate on the innovation) to detect a manoeuvre and inflate the covariance
- then an interacting-multiple-model filter: ballistic and manoeuvring hypotheses run in parallel, weighted by likelihood
- only then retrain the policy on the new estimate statistics, the same recipe as the 51→97 step

**Multiple agents** the shape I would build, not something I have built.

- centralised training, decentralised execution: one critic sees every agent during training; each policy acts on its own observations at run time
- a communications model in the simulator (what each agent can see of the others, with delay and dropout), because the policy can only use what it would really have
- the simulator is already vectorised over 1,024 independent episodes; making agents share an episode is the structural change

First reading: MAPPO (Yu et al. 2022); Bar-Shalom on IMM.

# A16 PPO details, as configured

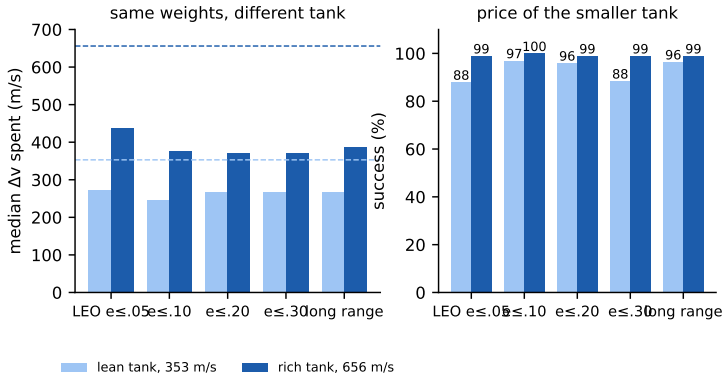
| knob | value |
|---|---|
| parallel environments | 1,024 |
| steps per stage | 50 M (mixture 200 M) |
| discount $\gamma$ | 0.995 |
| GAE $\lambda$ | 0.90 |
| clip ratio | 0.2 |
| entropy coefficient | 0.01 |
| value loss weight | 2.0 |
| epochs per batch | 1 |
| minibatch | 8,192 |
| BPTT horizon | 64 decisions |
| learning rate | 0.01 annealed; 0.001 consolidation |
| optimizer | Muon |
| reward clamp | $[-1, 1]$ before GAE |

**Reward** $+1$ success, $-1$ failure, 0 at the clock cap. Dense term: potential-based shaping $\Phi(s') - \Phi(s)$ on the mean-longitude gap and the orbit-shape mismatch; with $\gamma_s = 1$ it telescopes and cannot change the optimal policy.

**Why these** $\gamma = 0.995$ is a credit horizon of about 200 decisions; one epoch and a modest clip keep updates conservative with 1,024 correlated environments; entropy 0.01 kept the long coasts alive early (lower, and they died before the first success was sampled).

Never ablated: the 128 width, GELU, the LSTM itself. Library defaults, inherited rather than chosen.

# A17 Fuel awareness: the same weights spend less when they have less




Trained with the tank sampled per episode (353–656 m/s) and shown its fuel fraction, the generalist spends about 40 % less on a lean tank than on a rich one, for 1 to 11 points of success.

Against a linearised direct-burn reference the lean spend is 1.02–1.07×. A low-learning-rate consolidation stage also cut spend 14–21 % unprompted.